

Program : Diploma in Instrumentation Engineering	
Course Code : 4087	Course Title: Process Control
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To illustrate basic concept and operation of control valves.
- To design electronic converters and electronic controllers using op-amp.
- To demonstrate the various characteristics of control valve.
- To simulate various controller modes using MATLAB Simulink.
- To simulate various control system problems and study the response using MATLAB Simulink.

Course Prerequisites:

Topic	Course code	Course name	Semester
Computer knowledge		Introduction to IT System	1
Operational Amplifiers		Analog Circuits for Instrumentation	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Implement error detector, Instrumentation Amplifier, Voltage to Current converter, Current to Voltage converter. Different controller modes (P, PI, PD, PID). etc Design and setup alarm circuits for simple process control applications and apply these circuits in suitable situations	10	Applying

CO2	Demonstrate the various characteristics of control valves	10	Applying
CO3	Simulate various controller modes using MATLAB& Simulink.	12	Applying
CO4	Simulate various control system problems and study the response using MATLAB & Simulink.	10	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2	3						
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement analog Electronic Controllers and Converters		
M1.01	Design and setup Error detector circuit and Instrumentation amplifier using Op-amp.	2	Applying
M1.02	Design and setup current to voltage and voltage to current converters using OP amp.	2	Applying
M1.03	Design Simple alarm circuits using universal gates and comparator for Process control application	3	Applying
M1.04	Design and setup P, PI, PD and PID controllers using OP amp.	3	Applying
CO2	Demonstrate the various characteristics of control valves		
M2.01	Plot the curve between stem position and flow rate of a control valve.	3	Applying
M2.02	Plot the curve between control signal and flow rate of a control valve.	3	Applying
M2.03	Calculate the flow coefficient of control valve.	4	Applying
	Lab Test - II	2	

CO3	Simulate various controller modes using MATLAB & Simulink		
M3.01	Simulate Proportional and Proportional Integral controller using MATLAB & SIMULINK.	4	Applying
M3.02	Simulate Proportional derivative and Proportional Integral derivative controller using MATLAB & Simulink.	4	Applying
M3.03	Simulate instrumentation amplifier using MATLAB & Simulink.	4	Applying
CO4	Simulate various control system problems and study the response using MATLAB & Simulink.		
M4.01	Flow control process: plot the response curve for step response of setpoint change. Simulate for standard test signal	5	Applying
M4.02	Temperature control process: plot the response curve for step response of setpoint change. Simulate for standard test signal	5	Applying
	Lab Test - II	1	

Text / Reference:

T/R	Book Title/Author
R1	Curtis. D. Johnson, Process control instrumentation technology
R2	Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company
R3	Krishnakanth, Computer based Industrial Control
R4	Donald .P. Eckman, Automatic process control
R5	Ernest O Doebelin, Measurement Systems application and Design 5th Edition

Online Resources:

Sl.No	Website Link
1	https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf
2	https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/108108114/lec25.pdf
3	http://ctms.engin.umich.edu/CTMS/index.php?example=Introduction&section=Control Digital
4	https://www.mathworks.com/solutions/power-electronics-control.html